

Team Proposal

The purpose of the group proposal is to consider an engineering problem and propose a solution that involves both software and hardware on the “scale” of a capstone project. The proposal is all documentation, not coding or prototyping.* This is similar to a business proposal, in responding to a proposal request or generating new business opportunities.

1. Team and topic

You will be assigned to a group (nominal size three students) that will work together to create an engineering proposal. You can expect that your group may be a mix of majors (EE, CS and others), so you will need to use good communication skills, team goal-setting and incorporating the work of many “experts” from different domains. After your team is formed, you will need to meet and find a topic that uses the expertise of your team members. This can be challenging, but try to find a topic that will involve all of the team members. You can browse prior capstone projects for ideas at: <https://www.ceias.nau.edu/capstone/d4p/>

Your proposal topic will need to be approved soon after your group forms, so contact your instructor for approval as soon as possible.

2. Research

Your team will likely have to research various parts of the topic and find the problem and a proposed solution. You should have research widely, with at least six sources. Try to avoid textbooks, and use good primary (not secondary) sources. You should give preference to academic sources, but whitepapers are acceptable.

3. Outline

An engineering proposal should show your problem, a solution, and a suitable decomposition of the solution to subsystems. Your proposed solutions to the hardware and software aspects of the modular components should have well developed details: good outlines of algorithms, estimates of work and components required. Use graphics as needed to describe your solution, such as a page for a functional decomposition and a page for a Gantt chart.

As this is a multi-week assignment, you will need to plan your effort, and be prepared for briefing your progress and ideas to others (and incorporating feedback) during the time of proposal preparation.

You should have at least three sources in your proposal bibliography. Try to avoid textbooks, and use good academic database sources. You should give preference to academic sources, but whitepapers are acceptable.

4. Formatting

Your Group Proposal should meet the following formatting guidelines:

- At least five written pages (which excludes figures and the bibliography)
- MS Word document

- Font: Times New Roman, 12 point
- A header on the first page only with the title, your names, class, and date single spaced (1.15)
- Double spaced text = 2.0 line spacing in Word
- Indentation of new paragraphs by 0.5 inches
- “Normal” margins in Word, 1 inch
- The sections of your proposal should be: Problem, Research (or Background), Solution
- Graphics used as needed to clearly convey the proposal
- A bibliography of your sources in either ACM or IEEE format, in-text citations

Post your document to the “Proposal” assignment link. Do not post PDF documents, post only MS Word documents.

NOTE: You will also make a group presentation of this proposal.

* If your team selected to work with hardware, see additional guidance.